

1 Highlights

2 **Beyond Accuracy: A Validation Framework for Machine Learning** 3 **in Cryptocurrency Trading**

4 Jaewook Kim

- 5 • VALID: to our knowledge the first checklist-style protocol pairing binary
6 statistical and economic gates for financial ML
- 7 • Bull bias: tree-based crypto ML models predict 90–97% long without
8 class balancing
- 9 • Statistical significance does not guarantee economic profitability: nine
10 Bonferroni survivors all carry PBO=1.0
- 11 • Net Sharpe falls monotonically with trading frequency; every 15-minute
12 variant loses money after costs
- 13 • Monte Carlo ($n=200$): 27% AUC false positive rate, eliminated by
14 CPCV with PBO

Beyond Accuracy: A Validation Framework for Machine Learning in Cryptocurrency Trading

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Abstract

Machine learning strategies for cryptocurrency trading routinely report exceptional backtest results, yet practitioners consistently fail to replicate them. We identify three systematic failure modes—directional prediction bias, statistical-economic disconnect, and transaction cost omission—through 340 strategy variants across four timeframes and three cryptocurrency assets. To address these failures, we propose VALID (Validation Architecture for Learning-based Investment Decisions), a 12-item reporting and validation framework for financial ML research—to our knowledge the first checklist-style protocol for financial ML to pair binary statistical and economic gates in a fixed reporting order.

We demonstrate that gradient-boosted models predict long 90–97% of the time without class balancing; that a run passing the permutation test ($p=0$) can still fail combinatorial purged cross-validation and be economically irrelevant; and that net Sharpe falls monotonically with trading frequency. Monte Carlo analysis (200 iterations $\times$ 4 asset-timeframe settings) shows 27–30% false positive rates for AUC-based validation alone, reduced to 0% by CPCV with PBO. Applying five multiple-testing corrections to the corpus, nine variants survive Bonferroni—all with $PBO=1.0$ —and none survives the Deflated Sharpe Ratio. We report null-distribution analysis of $\text{Var}(\text{SR}_{\text{IS}})$ providing evidence consistent with Witzany’s (2021) PBO critique. A systematic audit of 80 papers reveals that the median study satisfies only 2.5 of 12 VALID items. We release an open-source implementation for reproducibility.

Keywords: financial machine learning, cryptocurrency trading, validation framework, backtest overfitting, reporting standard

JEL: G11, G14, G17, C45, C52

Revision note (v3.2, August 2026). This version corrects the previous draft: a cost table built on a synthetic placeholder signal has been replaced with corpus statistics; a case study that combined metrics from two pipeline runs has been rewritten from a single run’s artifacts; the parameter-flatness claim has been narrowed; the self-assessment was revised from 10/12 to 9/12. All corrections were identified by applying the VALID audit process to our own paper. Full changelog: https://github.com/orcajae/valid-framework/blob/main/CHANGELOG_CR.md

1. Introduction

The application of machine learning to cryptocurrency trading has grown rapidly, producing a literature characterized by strikingly positive results. Cross-sectional factor models report weekly long-short returns of 3.87% (Fieberg et al., 2025). LSTM ensembles achieve annualized Sharpe ratios exceeding 3.0 (Xu et al., 2022). Tree-based models demonstrate AUC scores above 0.80 on directional prediction tasks (Li et al., 2024). Taken at face value, these findings suggest ML-driven crypto trading offers among the highest risk-adjusted returns in any asset class.

Yet the gap between published results and practitioner experience is vast. In traditional finance, this disconnect is well documented: Ioannidis (2005) demonstrated that most published research findings are false under conditions of low pre-study odds and high researcher flexibility. Harvey et al. (2016) showed that the majority of 316 published equity factors are likely false discoveries, proposing a t -ratio threshold of 3.0 rather than the conventional 2.0. McLean and Pontiff (2016) documented 26% out-of-sample decay and 58% post-publication decay in equity anomalies. Hou et al. (2020) replicated 452 anomalies and found 65% fail—82% after multiple testing correction. The cryptocurrency ML literature has not yet undergone equivalent scrutiny.

Kapoor and Narayanan (2023) identified data leakage affecting 294 studies across 17 scientific fields, proposing a taxonomy of eight leakage types and a “model info sheet” reporting template. Their work, published in *Patterns* and subsequently extended to the REFORMS framework in *Science Advances* (Kapoor et al., 2024), demonstrated that domain-specific validation standards can substantially reduce false positive rates. However, no analogous standard exists for financial ML. TRIPOD+AI (Collins et al., 2024) covers clinical

prediction models. REFORMS addresses general ML-based science. MI-CLAIM (Norgeot et al., 2020) targets clinical AI. Not one published checklist addresses the unique challenges of ML-based trading strategy development: temporal data dependencies, transaction cost modeling, class imbalance in directional prediction, or the distinction between statistical and economic significance.

This gap has consequences. Without standardized validation, published crypto ML results vary wildly in methodological rigor. Some studies omit transaction costs entirely; others use unrealistically low cost assumptions. Few test for directional prediction bias. Even fewer apply advanced overfitting diagnostics such as Combinatorial Purged Cross-Validation (CPCV) or Probability of Backtest Overfitting (PBO), despite their demonstrated superiority over traditional walk-forward methods (Arian et al., 2024).

1.1. Contributions

This paper makes five contributions:

1. **The VALID framework.** We propose a 12-item validation and reporting checklist for financial ML research—to our knowledge the first checklist-style protocol for this field to pair binary statistical and economic gates in a fixed reporting order, complementing rather than superseding prior backtesting guidance.
2. **Systematic literature audit.** We survey 80 published cryptocurrency ML trading papers (2018–2026), coding each for validation methodology, cost assumptions, class balancing, baseline comparison, and code availability.
3. **Three empirical failure modes at scale.** Using 340 strategy variants across four timeframes and three cryptocurrency assets (BTC, ETH, SOL), we document: (a) bull bias, (b) statistical-economic disconnect, and (c) cost illusion.
4. **Monte Carlo false positive quantification.** We generate 200 synthetic price series per setting across four asset-timeframe configurations, demonstrating that AUC-based evaluation produces 27–30% false positives while CPCV + PBO reduces this to 0% [0%, 1.9%] in every setting. Ablation analysis identifies CPCV/PBO (Item V4) as the single most impactful validation component.
5. **Multiple-testing evidence.** Applying five correction methods to the corpus, nine variants survive Bonferroni—every one with PBO = 1.0—while the Deflated Sharpe Ratio rejects all 340. We also derive a Monte

110 Carlo null distribution of $\text{Var}(\text{SR}_{\text{IS}})$ (real 0.012, null 95th percentile
 111 0.328) and show that the absolute and null-relative flatness criteria
 112 disagree, evidence consistent with Witzany’s (2021) PBO critique.

113 1.2. Notation and Abbreviations

114 Table 1 defines the key terms used throughout this paper.

Table 1: Key abbreviations and definitions.

Term	Definition
AUC	Area Under the ROC Curve; classification performance metric
CPCV	Combinatorial Purged Cross-Validation; generates $\binom{N}{k}$ temporal train-test splits
CUSUM	Cumulative Sum filter; event sampling method for non-uniform time series
FPR	False Positive Rate; proportion of null (signal-free) datasets incorrectly classified as positive
PBO	Probability of Backtest Overfitting; fraction of CPCV paths where in-sample-best underperforms out-of-sample
SR	Sharpe Ratio; annualized risk-adjusted return
TB	Triple Barrier labeling; assigns $\{-1, 0, +1\}$ labels based on profit/stop-loss/time barriers
$\text{Var}(\text{SR}_{\text{IS}})$	Variance of in-sample Sharpe ratios across configurations; measures parameter-landscape diversity
VALID	Validation Architecture for Learning-based Investment Decisions

115 1.3. Paper Organization

116 Section 2 reviews related work. Section 3 presents the systematic literature
 117 audit. Section 4 introduces the VALID framework. Sections 5 and 6 provide
 118 empirical validation. Section 7 discusses implications. Section 8 concludes.

119 2. Related Work

120 2.1. Machine Learning for Cryptocurrency Trading

121 The premise that ML can profitably trade cryptocurrency rests on market
 122 inefficiency; Urquhart (2016) provided early evidence that Bitcoin prices do

not follow a random walk, though subsequent work suggests efficiency has increased over time. The crypto ML literature has grown rapidly since 2019. Fang et al. (2022) surveyed 146 papers and identified technical analysis, fundamental analysis, and ML-based approaches as the three dominant paradigms, noting that approximately half of surveyed studies omit transaction costs. Early work focused on price direction prediction: McNally et al. (2018) applied LSTM and Bayesian-optimized RNNs to Bitcoin, while Alessandretti et al. (2018) tested gradient-boosted methods across multiple cryptocurrencies. Sun et al. (2020) proposed a LightGBM-based forecasting model, and Vo and Yuen (2020) explored multi-step prediction with deep learning. Sebastião and Godinho (2021) examined how changing market conditions affect ML trading performance. More recent work has expanded to cross-sectional factor models: Fieberg et al. (2025) developed the CTREND factor across 3,244 coins, while Cakici et al. (2024) applied 12 ML models to 37 cryptocurrency-specific factors. Leung and Zhao (2021) provide a comprehensive overview of cryptocurrency trading and exchange mechanics.

The methods adopted in crypto trading mirror those validated in traditional equities. Gu et al. (2020) demonstrated that gradient-boosted trees and neural networks dominate equity return prediction, and these same architectures now constitute the majority of crypto ML studies. Krauss et al. (2017) showed that ensemble methods achieve statistical arbitrage on the S&P 500 but noted the critical role of transaction costs in eroding gross alpha—a finding we extend to crypto in Section 5. Deep learning surveys (Olorunnimbe and Viktor, 2023; Siami-Namini et al., 2019; Jiang et al., 2024) document rapid LSTM adoption, yet neither addresses the class imbalance problem we identify. Beyond supervised learning, Borrageiro et al. (2022) applied reinforcement learning to FX trading, and Zhang et al. (2025) review neural algorithmic trading systems; both paradigms face the same validation gaps that VALID addresses. Schnaubelt (2022) applied DRL to cryptocurrency limit order placement, reporting profitable results but without CPCV or PBO validation.

However, several concerns have emerged. Jaquart et al. (2021) explicitly warned that imbalanced training sets may cause classifiers to predict the majority class regardless of input features. Lahmiri and Bekiros (2019) found that naive models consistently outperform ML and deep learning in univariate crypto forecasting. Grądzki et al. (2025) demonstrated that information-driven bars with triple barrier labeling improve upon standard approaches, but their cost assumptions (10 basis points) may understate realistic execution costs.

161 More broadly, Zhu et al. (2023) documented common pitfalls in ML-based
162 science—including data leakage, overfitting, and inadequate baselines—that
163 map directly onto the failure modes we quantify.

164 *2.2. Backtest Overfitting and Validation Methodology*

165 White (2000) proposed the Reality Check for data snooping, later refined
166 by Hansen’s (2005) Superior Predictive Ability test. Bailey and López de
167 Prado (2014); Bailey et al. (2017) introduced PBO and DSR as tools for
168 detecting backtest overfitting. Arian et al. (2024), published in Knowledge-
169 Based Systems, compared CPCV against walk-forward methods, introducing
170 Bagged CPCV and Adaptive CPCV variants.

171 Witzany (2021) provided the only formal critique of PBO, demonstrating
172 that CSCV/PBO exhibits negative bias when strategies have similar returns:
173 “the best IS model tends to the worst OOS not because of the models but due to
174 the design of the method itself.” Bailey et al. (2017) themselves acknowledged
175 that low PBO does not guarantee positive out-of-sample performance. The
176 Sharpe ratio itself requires careful statistical treatment (Lo, 2002), particularly
177 when comparing strategies across different holding periods.

178 *2.3. Statistical Versus Economic Significance*

179 Harvey et al. (2016) argued that extensive data mining requires t -ratios
180 above 3.0. Novy-Marx and Velikov (2016) demonstrated that most equity
181 anomalies with monthly turnover above 50% become unprofitable after costs.
182 Blitz et al. (2023) showed that even short-term alpha signals in equities are
183 substantially eroded by transaction costs. Patton and Weller (2020) found
184 momentum implementation costs of 7.2–7.6% annually eliminate most profits.
185 Chen and Velikov (2023) showed that post-publication decay combined with
186 trading costs destroys 93% of anomaly returns.

187 In cryptocurrency markets, Makarov and Schoar (2020) documented signif-
188 icant cross-exchange price differences. Kim et al. (2021) developed a volatility
189 index for cryptocurrencies, highlighting the extreme volatility regime that
190 amplifies transaction cost impact. Almeida and Gonçalves (2024) system-
191 atically reviewed crypto market microstructure, identifying bid-ask spreads
192 of 2–50+ basis points. No published study has systematically quantified
193 the gross-to-net alpha gap across multiple timeframes for ML-based crypto
194 strategies.

195 2.4. Reporting Standards for ML-Based Research

196 TRIPOD (Collins et al., 2015) provides 22 items for clinical prediction
197 model reporting and has accumulated over 4,000 citations. TRIPOD+AI
198 (Collins et al., 2024) extends this with 27 items. REFORMS (Kapoor et al.,
199 2024) proposes 32 items for general ML-based science. MI-CLAIM (Norgeot
200 et al., 2020) provides a 6-step checklist for clinical AI. Model Cards (Mitchell
201 et al., 2019) propose documentation standards for trained ML models.

202 Critically, **no published reporting standard addresses financial**
203 **ML or algorithmic trading**. This gap motivates the VALID framework
204 proposed in Section 4.

205 2.5. Cryptocurrency Momentum and Simple Benchmarks

206 Time-series momentum in Bitcoin was established by Liu and Tsyvinski
207 (2021). Yang (2025) applied risk-managed momentum to crypto, reporting
208 Sharpe improvements from 1.12 to 1.42. Grobys et al. (2025) documented that
209 crypto momentum is subject to severe crashes mitigable through volatility
210 management.

211 3. Systematic Literature Audit

212 3.1. Search Strategy and Paper Selection

213 We searched Google Scholar, Scopus, and Web of Science for papers
214 published between 2018 and 2026 using the query terms “cryptocurrency” AND
215 (“machine learning” OR “deep learning”) AND (“trading” OR “prediction”).
216 After screening, we retained 80 papers for full-text analysis, of which 74
217 are empirical crypto ML studies. We classify a paper as empirical if it
218 applies a machine learning method to real cryptocurrency market data for
219 a trading or prediction task; surveys, synthetic-only methodological studies,
220 and non-crypto or non-ML studies are excluded (six papers).

221 3.2. Coding Methodology

222 Each paper was coded on seven dimensions corresponding to the VALID
223 framework’s core items (Table 2).

Table 2: Literature audit coding dimensions.

Dimension	Coding Question
D1: Cost modeling	Are transaction costs included?
D2: Class balance	Is directional prediction bias addressed?
D3: Temporal split	Is train/test splitting temporal or random?
D4: Validation	Walk-forward, CPCV, k-fold, or holdout?
D5: Baselines	Compared against buy-and-hold? Simple rules?
D6: Net performance	Is cost-adjusted performance reported?
D7: Reproducibility	Is code or data publicly available?

Table 3: Literature audit results ($n = 74$ empirical papers).

Dimension	Failing	Rate	95% CI	VALID Item
D1: Costs omitted	40/74	54%	[43%, 65%]	V7
D2: No class balance	54/74	73%	[62%, 82%]	V1, V2
D3: Random split	5/74	7%	[3%, 15%]	V3
D4: Weak validation	15/74	20%	[13%, 31%]	V4
D4: CPCV used	0/74	0%	[0%, 5%]	V4
D5: BnH only baseline	26/74	35%	[25%, 46%]	V9
D6: No net performance	40/74	54%	[43%, 65%]	V7, V8
D7: No code available	63/74	85%	[75%, 91%]	V12

224 3.3. Audit Results

225 Table 3 summarizes the findings across 74 empirical papers, with 95%
226 Wilson confidence intervals.

227 Class imbalance is the most neglected dimension: 73% [62%, 82%] of
228 papers do not address class balance. Cost omission remains widespread at
229 54% [43%, 65%]. CPCV adoption is negligible: no empirical paper in our
230 sample uses it. Reproducibility is poor: 85% [75%, 91%] provide neither
231 code nor data. The median paper satisfies only 2.5 of 12 VALID items (see
232 Figure 1).

233 4. The VALID Framework

234 4.1. Design Principles

235 The VALID (Validation Architecture for Learning-based Investment De-
236 cisions) framework follows three principles: (1) finance-specific, addressing
237 pitfalls beyond REFORMS/TRIPOD; (2) evidence-based, motivated by quan-
238 tified failure modes; (3) actionable, specifying what to report and how to
239 test.

240 4.2. The 12 VALID Items

241 Table 4 presents the 12 items.

242 4.2.1. Item Descriptions

243 **V1: Report prediction class distribution.** Authors must report the
244 proportion of long, short, and neutral labels in both training and test sets.
245 In trending cryptocurrency markets, base rates of 60–80% long are common,
246 making majority-class memorization a viable strategy for any classifier. With-
247 out this disclosure, reviewers cannot distinguish genuine predictive power
248 from base-rate exploitation.

249 **V2: Test with and without class balancing.** Results should be
250 reported under both balanced and unbalanced conditions. As demonstrated
251 in Section 5 (Failure Mode 1), class balancing eliminates directional bias but
252 typically does not improve AUC, revealing whether the model has learned
253 signal or structure. Acceptable balancing methods include sample weights,
254 SMOTE, and undersampling; the chosen method should be stated explicitly.

255 **V3: Use temporal splitting only.** All train-test splits must respect
256 chronological order, with purge and embargo periods to prevent information
257 leakage from overlapping feature lookback windows. Random or stratified

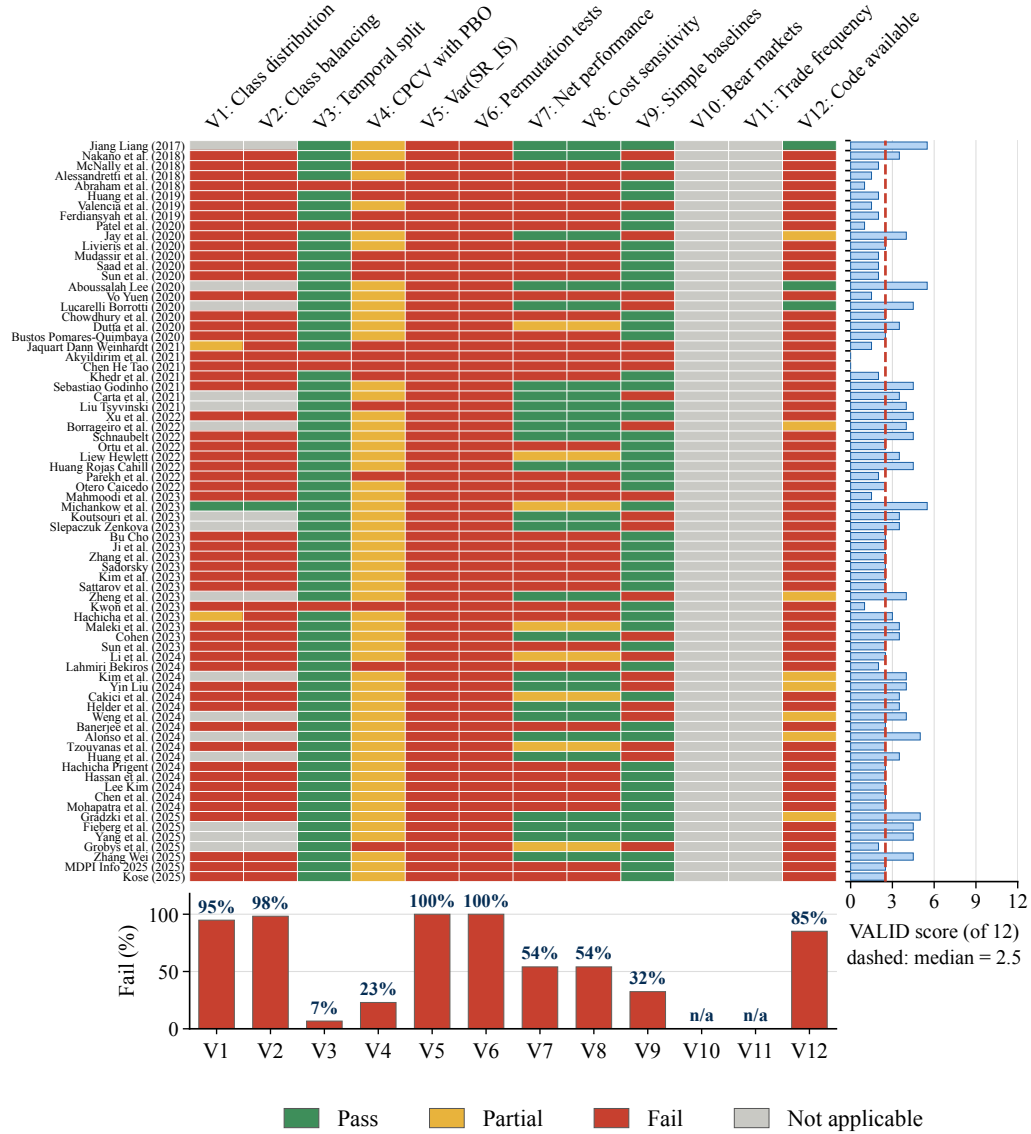


Figure 1: VALID compliance across 74 empirical cryptocurrency ML papers (2017–2025). Each row represents one paper (chronologically ordered); columns represent VALID items V1–V12. The median paper satisfies 2.5 of 12 items. Per-paper states are derived from the coding sheet (Table 2) under the dimension-to-item mapping printed in Table 3. The failure rates in the lower panel use applicable-only denominators and so differ from Table 3, which divides by all 74: V1 and V2 fail in 95% and 98% of the 58 papers where a directional class balance applies, and in 73% of all 74. V5 and V6 were not coded per paper and are recorded as universal non-compliance; V10 and V11 were not directly assessable from published text and are coded as N/A.

Table 4: The 12 VALID items.

#	Item	Rationale	Failure Mode
V1	Report prediction class distribution	Base-rate learning on trending assets	Bull bias
V2	Test with/without class balancing	Structural class imbalance in crypto	Bull bias
V3	Use temporal splitting only	Information leakage across time, including feature lookback windows	Temporal leakage
V4	Apply CPCV with PBO	Holdout insufficient for large search	Overfitting
V5	Report $\text{Var}(\text{SR}_{\text{IS}})$	$\text{PBO} = 0$ from flat landscape (Witzany, 2021)	PBO misinterpretation
V6	Include permutation tests (≥ 100)	Confirm signal existence	Spurious patterns
V7	Report net performance with costs	Cost omission inflates alpha	Cost illusion
V8	Cost sensitivity analysis	Cost assumptions can flip alpha sign	Cost illusion
V9	Compare against simple baselines	Weak baselines inflate relative performance	Weak baselines
V10	Evaluate in bear markets and regime transitions	Bull-market trend-following not informative	Regime overfit
V11	Report trade frequency	High-frequency cost headwinds	Hidden turnover
V12	Provide code for reproducibility	$< 10\%$ of papers provide code	Irreproducibility

258 k-fold splitting violates the temporal dependence structure of financial data
 259 and produces upward-biased performance estimates. Our audit finds 7% of
 260 papers still use random splitting (Table 3).

261 **V4: Apply CPCV with PBO.** Combinatorial Purged Cross-Validation
 262 generates $\binom{N}{k}$ train-test combinations from N chronological groups, producing
 263 multiple non-overlapping backtest paths (see Equation 4). PBO quantifies
 264 the probability that the in-sample-best configuration underperforms out-of-
 265 sample. Our ablation analysis (Table 12) shows that removing V4 alone
 266 increases the false positive rate from 0% to 27%, making it the single most
 267 impactful validation component.

268 **V5: Report parameter-space variance $\text{Var}(\text{SR}_{\text{IS}})$.** A low PBO can
 269 be misleading when all strategy configurations perform similarly in-sample,
 270 so that the search cannot distinguish between them (Equation 5). Authors
 271 should compute $\text{Var}(\text{SR}_{\text{IS}})$ across all tested configurations and compare it
 272 against a null distribution derived from synthetic data. If the observed
 273 variance falls below the 5th percentile of the null, PBO should be flagged as
 274 uninformative rather than reassuring (Table 9).

275 **V6: Include permutation tests (≥ 100 iterations).** Permutation
 276 testing establishes whether a model’s performance exceeds what would be
 277 expected from random label assignment. We recommend a minimum of 100
 278 permutation iterations and reporting the resulting p-value alongside standard
 279 metrics. This provides a complementary check to CPCV/PBO by testing for
 280 signal existence rather than overfitting.

281 **V7: Report net performance with realistic transaction costs.**
 282 Gross Sharpe ratios are meaningless for deployment decisions. Authors must
 283 report net performance after round-trip transaction costs including exchange
 284 fees, estimated slippage, and funding rates where applicable. Our analysis
 285 shows net Sharpe falling monotonically with trading frequency (Table 10)
 286 and, in the one run where gross and net are both recorded, 82% of the gross
 287 Sharpe consumed at 18 bp.

288 **V8: Conduct cost sensitivity analysis.** Because true execution costs
 289 vary by venue, time, and order size, authors should report performance across
 290 at least three cost levels (e.g., 0, 18, and 50 basis points round-trip). A strategy
 291 whose alpha flips sign within realistic cost bounds has limited practical value.
 292 This analysis also reveals the cost breakeven point—the maximum tolerable
 293 transaction cost for positive net returns.

294 **V9: Compare against simple baselines beyond buy-and-hold.** Buy-
 295 and-hold is a necessary but insufficient baseline. Authors should include at

least one rule-based alternative with comparable trading frequency (Table 11). In our experiments, RSI and MACD crossover—strategies implementable in under 10 lines of code—outperform the best ML variant at matched frequency.

V10: Evaluate in bear markets and regime transitions. Strategies trained predominantly on bull-market data may perform well on average but fail catastrophically during drawdowns. Authors should report performance conditional on market regime (e.g., calm vs. stressed, as in Figure 5). A strategy that underperforms buy-and-hold during bear markets offers no hedging value despite positive aggregate statistics.

V11: Report trade frequency and turnover. High-frequency strategies incur proportionally higher transaction costs, creating a hidden drag that aggregate Sharpe ratios obscure. Reporting annualized trade count enables reviewers to assess whether reported returns survive realistic execution. In our sample, every 15-minute variant shows $SR \leq -1.33$ after costs.

V12: Provide code and data for reproducibility. Our audit finds 85% of papers provide neither code nor data (Table 3). Without reproducibility artifacts, claimed results cannot be independently verified. We recommend releasing, at minimum, the training pipeline, evaluation code, and a sample dataset or synthetic data generator sufficient to reproduce the main results.

4.3. Relationship to Existing Standards

VALID supplements REFORMS with finance-specific items (V7, V8, V11 for cost modeling; V1, V2 for class balance) absent from all existing frameworks. Table 5 highlights the coverage gap.

Table 5: VALID versus existing ML reporting standards.

Standard	Domain	Items	Cost Model	Class Bal.
TRIPOD+AI	Clinical ML	27	N/A	No
REFORMS	General ML	32	No	Partial
MI-CLAIM	Clinical AI	6	N/A	No
NeurIPS Checklist	ML research	15	No	No
VALID	Financial ML	12	Yes	Yes

4.4. Application Protocol

We recommend two stages: **Stage 1 (Reporting)**: V1–V6, V9, V12; **Stage 2 (Deployment)**: V7, V8, V10, V11. A strategy passing Stage 1 but

322 failing Stage 2 has scientific value but limited practical value—a distinction we
 323 term the **statistical-economic disconnect**. Figure 2 provides an overview
 324 of the VALID framework architecture and its relationship to the three failure
 325 modes documented in Section 5.

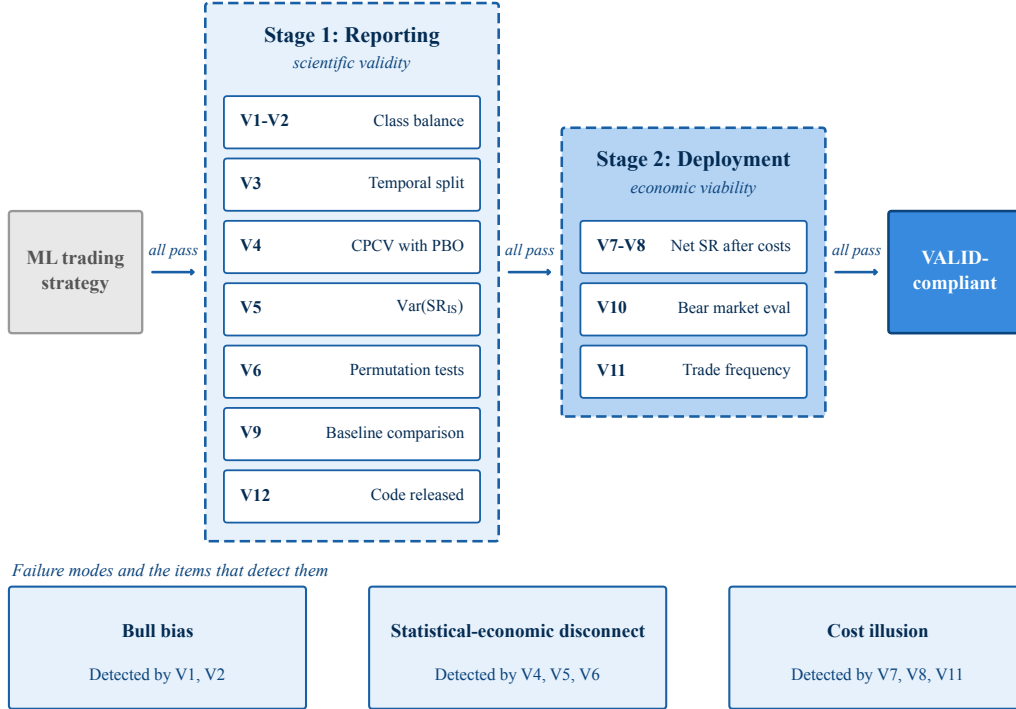


Figure 2: VALID framework overview. Stage 1 (Reporting) items V1–V6, V9, V12 address scientific validity; Stage 2 (Deployment) items V7, V8, V10, V11 address economic viability. Three empirical failure modes are detected by specific item subsets: bull bias (V1, V2), statistical-economic disconnect (V4–V6), and cost illusion (V7, V8, V11). A strategy passing Stage 1 but failing Stage 2 exhibits the statistical-economic disconnect.

326 5. Empirical Validation: Three Failure Modes

327 5.1. Methodology and Notation

328 We deliberately employ standard, well-documented model architectures
 329 rather than novel ones, as our objective is to evaluate validation methodology,
 330 not to propose new prediction algorithms. This subsection defines the key
 331 methods used throughout our experiments.

332 5.1.1. Triple Barrier Labeling

333 Following López de Prado (2018), each observation is assigned a label
 334 $y_i \in \{-1, 0, +1\}$ based on the first barrier touched by the price path after
 335 event time t_i . Let p_t denote the price at time t . Three barriers are defined:

$$\text{Upper: } p_{t_i+\tau} \geq p_{t_i}(1 + \theta_u) \quad (1)$$

$$\text{Lower: } p_{t_i+\tau} \leq p_{t_i}(1 - \theta_l) \quad (2)$$

$$\text{Vertical: } \tau = T_{\max} \quad (3)$$

336 where $\tau > 0$ is the elapsed time since the event, so that $p_{t_i+\tau}$ is the price
 337 τ periods after t_i and the label is set by whichever of the three conditions
 338 holds at the smallest τ ; $\theta_u, \theta_l > 0$ are profit-taking and stop-loss thresholds
 339 scaled to $\hat{\sigma}$, the exponentially weighted estimate of local return volatility at t_i
 340 ($\theta_u = 2.0\hat{\sigma}$, $\theta_l = 2.5\hat{\sigma}$ in our daily configuration; Equations 1–3); and $T_{\max}$ is
 341 the maximum holding period. Events are sampled via CUSUM filtering on
 342 cumulative log-returns with threshold h .

343 5.1.2. Combinatorial Purged Cross-Validation (CPCV)

344 Given N chronologically ordered groups and a test size of k groups,
 345 CPCV generates $\binom{N}{k}$ unique train-test combinations, each respecting temporal
 346 ordering with purge and embargo periods to prevent information leakage.
 347 In our configuration ($N = 6$, $k = 2$), this yields $\binom{6}{2} = 15$ combinations,
 348 producing $\phi = (k/N)\binom{N}{k} = 5$ non-overlapping backtest paths. For each
 349 combination j , a model is trained on the $N - k$ training groups and evaluated
 350 on the k test groups, producing an out-of-sample Sharpe ratio $\text{SR}_{\text{OOS}}^{(j)}$.

351 5.1.3. Probability of Backtest Overfitting (PBO)

352 Let M denote the number of candidate configurations evaluated in the
 353 search. For each CPCV path, the in-sample-best configuration is identified
 354 and its out-of-sample rank $\omega^{(j)} \in \{1, \dots, M\}$ is recorded. The relative rank is
 355 normalized as $\bar{\omega}^{(j)} = \omega^{(j)} / (M + 1)$ to avoid boundary values. PBO is defined
 356 as:

$$\text{PBO} = \frac{1}{\binom{N}{k}} \sum_{j=1}^{\binom{N}{k}} \mathbf{1}[\text{logit}(\bar{\omega}^{(j)}) \leq 0] \quad (4)$$

357 where $\text{logit}(x) = \ln(x/(1 - x))$. PBO ≈ 0 indicates that in-sample-best config-
 358 urations tend to rank in the top half out-of-sample; PBO ≈ 1 indicates severe
 359 overfitting (Equation 4; Bailey et al. 2017).

360 5.1.4. Parameter-Space Variance

361 To diagnose how much configurations differ in-sample (Witzany, 2021),
 362 we compute the variance of in-sample Sharpe ratios across all tested configu-
 363 rations:

$$\text{Var}(\text{SR}_{\text{IS}}) = \frac{1}{M-1} \sum_{m=1}^M \left(\text{SR}_{\text{IS}}^{(m)} - \overline{\text{SR}}_{\text{IS}} \right)^2 \quad (5)$$

364 where $\text{SR}_{\text{IS}}^{(m)}$ is the in-sample Sharpe ratio of configuration m and $\overline{\text{SR}}_{\text{IS}}$ their
 365 mean over the M configurations. Low $\text{Var}(\text{SR}_{\text{IS}})$ relative to a null distribution
 366 (Equation 5) indicates that all configurations perform similarly, rendering
 367 $\text{PBO} = 0$ uninformative rather than reassuring.

368 5.2. Strategy Universe

369 Our 340 strategy variants¹ are constructed as shown in Table 6:

Table 6: Composition of the 340-variant strategy universe.

Component	Description	Count
(a)	SHAP-derived rule combinations (applied to momentum benchmark)	64
(b)	Order flow features (individual + multi-asset)	55
(c)	External/on-chain/macro feature sets	45
(d)	ML pipeline: 3 assets $\times$ 4 TF $\times$ 3 tree families	36
(e)	Triple Barrier parameter sensitivity	36
(f)	Cross-section momentum configurations	24
(g)	Cost sensitivity & traditional baselines	24
(h)	Deep learning: 3 assets $\times$ 2 TF $\times$ 5 arch. (Transformer, LSTM, TCN, MLP $\times$ 2)	30
(i)	VALID ablation configurations	12
(j)	Volatility prediction overlays	8
(k)	Bull bias analysis (tree + DL)	6
Total unique configurations		340

370 Component (a) uses SHAP feature attributions (Lundberg and Lee, 2017)
 371 to derive interpretable trading rules from gradient-boosted models. Com-
 372 ponents (a)–(k) span three cryptocurrency assets (BTC, ETH, SOL), four
 373 timeframes (15-minute to daily), and eight model families (CatBoost, Light-
 374 GBM, Random Forest, Transformer, bidirectional LSTM, TCN, MLP 2-layer,

¹The corpus includes 18 cost-level entries (a rule-based benchmark and two signal proxies evaluated at six cost levels each). Excluding them, 55.0% of the remaining 322 variants are negative net of costs and 11 exceed the benchmark, one of which is the benchmark’s own entry in the baseline block; all corpus-level conclusions are unchanged.

MLP 3-layer). Bull bias analysis (component k) additionally includes unidirectional LSTM and SimpleRNN. Each was evaluated with 18 bp round-trip costs. Apart from the 18 cost-level entries described in footnote 1, all variants were executed through the full ML pipeline.

5.3. Failure Mode 1: Bull Bias

5.3.1. Experimental Setup

We train gradient-boosted tree models (CatBoost (Prokhorenkova et al., 2018), LightGBM (Ke et al., 2017), Random Forest) and deep learning models (2-layer LSTM, SimpleRNN) on hourly OHLCV data for BTC, ETH, and SOL with Triple Barrier labeling and CUSUM filtering (López de Prado, 2018). All tree-based methods follow the gradient boosting framework of Friedman (2001), with CatBoost’s ordered boosting (Dorogush et al., 2018) and XGBoost’s regularized objective (Chen and Guestrin, 2016) representing the two dominant implementations in recent crypto ML studies.

5.3.2. Results

Table 7 presents the results.

Table 7: Bull bias across assets and model families.

Asset (Model)	Unbal. Long%	Bal. Long%	AUC (u)	AUC (b)	Net SR
BTC (CatBoost)	97.2%	42.3%	0.502	0.501	−0.877
BTC (LSTM)	57.7%	52.3%	0.518	0.519	−1.149
BTC (SimpleRNN)	63.3%	—	—	0.520	—
ETH (CatBoost)	90.5%	45.2%	0.501	0.493	−1.283
SOL (CatBoost)	97.0%	32.2%	0.533	0.511	−0.206

Tree-based models exhibit extreme long bias (90–97%) while deep learning shows moderate-to-strong bias (58–86% across seven architectures including Transformer, bidirectional LSTM, and TCN). AUC is indistinguishable from random (0.50–0.52) across all model families. Bull bias extends the observation of Jaquart et al. (2021) to a systematic, multi-asset, multi-model phenomenon (see Figure 3).

5.3.3. Extended Multi-Asset Analysis

To confirm that bull bias is not an artifact of a single asset, timeframe, or model family, we extend the evaluation to 18 asset-timeframe-model

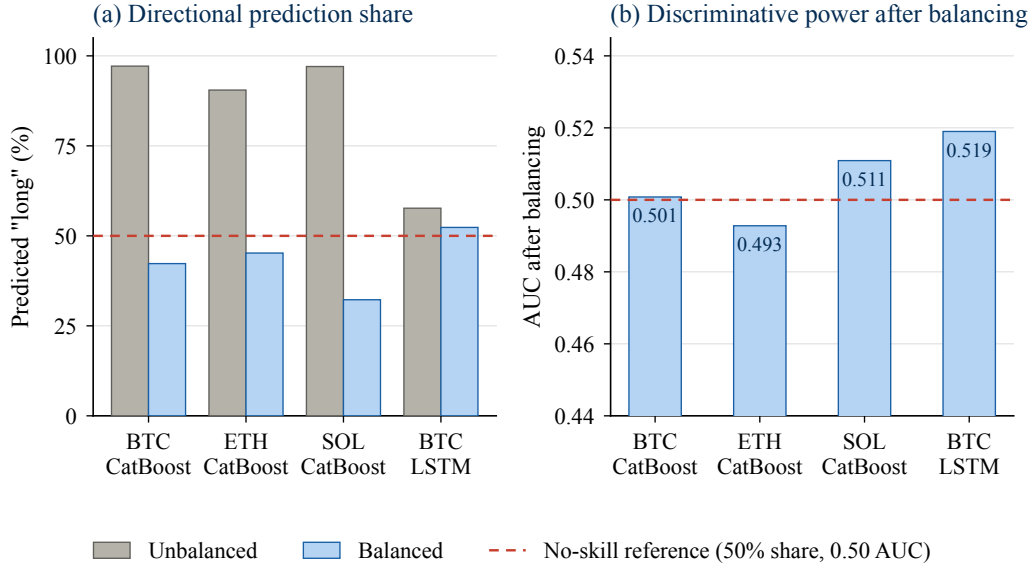


Figure 3: Bull bias, computed from the 340-variant corpus (the same rows as Table 7). (a) Share of test observations predicted “long” without and with class balancing: tree models fall from 90–97% to 32–45%, the LSTM from 57.7% to 52.3%. (b) AUC after balancing sits at chance level (0.493–0.519) for every model.

Table 8: Extended multi-asset bull bias (18 combinations).

Asset	Mean Unbal. Long%	Mean Bal. Long%	Mean AUC (bal)	Mean Net SR
BTC	85%	47%	0.484	−0.174
ETH	83%	52%	0.563	+0.023
SOL	80%	45%	0.542	+0.441

400 combinations: 3 assets $\times$ 2 timeframes (1-hour, daily) $\times$ 3 model families
 401 (CatBoost, LightGBM, Random Forest). Table 8 summarizes the results.

402 All assets exhibit unbalanced long ratios of 80–97% across all models and
 403 timeframes. Critically, class balancing eliminates directional bias but does
 404 not improve predictive power: balanced AUC remains indistinguishable from
 405 random (0.48–0.56) across all 18 asset-model combinations, and mean net SR
 406 is negative for BTC (−0.174). All 1-hour models receive PBO = 1.000. The
 407 phenomenon is timeframe-invariant and asset-invariant.

408 5.4. Failure Mode 2: Statistical-Economic Disconnect

409 5.4.1. The Disconnect

410 The order-flow 1h CatBoost run illustrates why Stage 1 requires more
 411 than significance testing. The model passes the permutation test decisively
 412 ($p = 0.000$; AUC 0.570 vs. a shuffled 95th percentile of 0.516): genuine
 413 signal exists. Its net SR of +0.135 against a benchmark of 0.917 already
 414 signals economic irrelevance—but a reviewer applying only significance tests
 415 would not see this coming. CPCV does: the same run’s PBO is 0.267,
 416 failing V4’s threshold. Selection on in-sample performance is not merely
 417 uninformative here but actively misleading: across CPCV configurations, the
 418 Spearman correlation between IS and OOS Sharpe is $\rho = -0.83$ ($p = 0.005$,
 419 $n = 9$ configurations), corroborated by the walk-forward analysis ($\rho = -0.73$,
 420 $n = 26$). Both statistics are configuration-level; we do not report a fold-level
 421 correlation. This aligns with Harvey et al. (2016) and Novy-Marx and Velikov
 422 (2016) on statistical-economic gaps (see Figure 4).

423 5.4.2. Parameter-Space Variance: Evidence Consistent with Witzany (2021)

424 Witzany (2021) demonstrated theoretically that CSCV/PBO exhibits neg-
 425 ative bias when strategies have similar returns. Arian et al. (2024) compared
 426 CPCV variants but did not measure $\text{Var}(\text{SR}_{\text{IS}})$ under null conditions.

427 We address this gap using 200-iteration Monte Carlo simulation (Section 6).
 428 Table 9 compares the real and null distributions:

429 The real data’s parameter-space variance falls below the Monte Carlo null
 430 distribution, but it exceeds the absolute flatness threshold of 0.01 that the
 431 same computation applies. Here the two flatness criteria disagree (absolute
 432 threshold 0.01 vs. MC null 95th percentile 0.328), illustrating why V5 requires
 433 the null to be pre-specified rather than derived after the fact. This provides
 434 empirical evidence consistent with Witzany’s analysis of PBO under similar

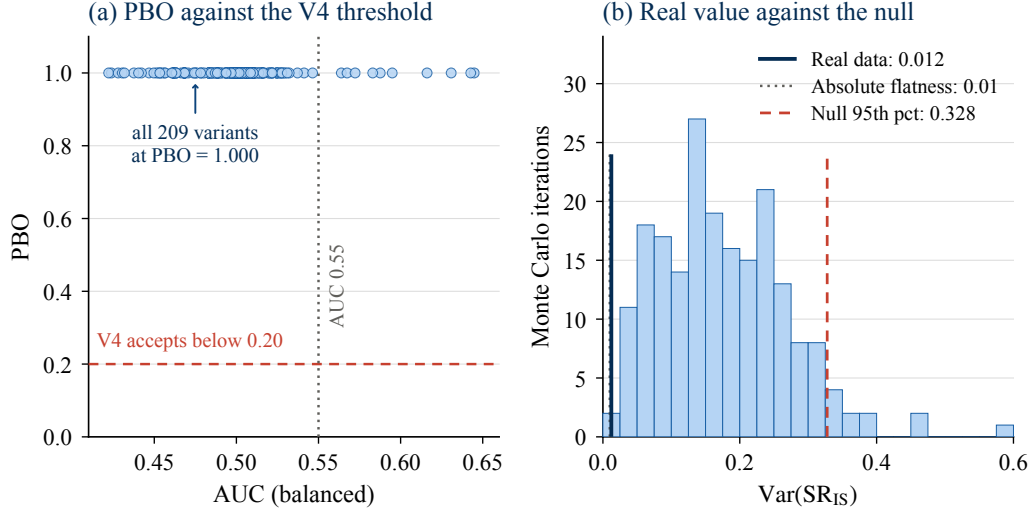


Figure 4: The PBO paradox. (a) PBO against AUC for the 209 corpus variants for which PBO was computed. Every one receives PBO of exactly 1.000—the ceiling, and five times the 0.20 below which V4 accepts—while AUC clusters at chance. (b) Parameter-space variance: the real data’s $\text{Var}(\text{SR}_{\text{IS}})$ of 0.012 falls below the Monte Carlo null distribution ($n = 200$, 95th percentile 0.328) while remaining above the absolute flatness threshold of 0.01—the two criteria disagree (Section 5.4.2).

Table 9: Parameter-space variance: real vs. null across four settings. [†]The real-data figure is computed on the 1-hour balanced pipeline (the base configuration without order-flow features), as the mean within-fold variance of in-sample Sharpe across model configurations.

Setting	Real Data	Null Mean	Null 95th pctl
BTC 1h [†]	0.012	0.176	0.328
ETH 1h	—	0.262	0.534
SOL 1h	—	0.296	0.617
BTC daily	—	0.176	0.328

strategy returns; it is not a verdict that our own parameter landscape is flat. An analytically derived acceptance threshold remains open.

5.4.3. Monte Carlo Confirmation

Monte Carlo analysis (Section 6) confirms these findings on synthetic data: AUC-based validation alone produces 27% false positives, while PBO eliminates all false positives (0% FPR, upper bound 1.9%). Full results are reported in Table 13.²

5.5. Failure Mode 3: Cost Illusion

5.5.1. Cross-Timeframe Cost Analysis

Table 10 reports net Sharpe ratios at 18 bp for the model variants of the corpus, grouped by trading frequency.

Table 10: Net Sharpe ratio (18 bp round-trip) of model variants in the 340-variant corpus, by trading frequency. Benchmark (DM): 0.917.

Frequency	n	Median	Best	Share < 0
15-minute	9	-1.80	-1.33	100%
1-hour	133	-1.32	+0.32	96%
4-hour	9	+0.28	+0.57	44%
Daily	77	+0.54	+1.98	22%

Model variants only; cost-level and rule-baseline rows of the corpus are excluded (Table 11).

Net performance degrades monotonically with trading frequency. The median 15-minute variant returns $SR = -1.80$ and all nine lose money after costs, whereas the median daily variant returns +0.54. Only daily-frequency variants exceed the DM benchmark at all (best 1.98); the best hourly variant reaches 0.32. In the one run where gross and net Sharpe are directly comparable, transaction costs consume 82% of the gross Sharpe ratio ($0.750 \rightarrow 0.135$) at 101 trades per year.

Our 18 bp fixed round-trip cost is conservative for hourly and daily frequencies (Binance taker fee: 10 bp + estimated slippage of 5–10 bp;

²Permutation test results are based on the first 100 iterations ($n = 100$); all other criteria are evaluated across all 200 iterations.

Almeida and Gonçalves 2024). For sub-hourly frequencies, volume-dependent slippage models (Almgren and Chriss, 2001) would further erode returns, strengthening rather than weakening our conclusions.

5.5.2. Comparison with Traditional Baselines

To ensure fair comparison across different trading frequencies, we group baselines by rebalancing frequency (Table 11) and compare each ML variant against a frequency-matched alternative.

Table 11: Strategy comparison by frequency band (18 bp round-trip costs).

Frequency	Strategy	CAGR	MDD	SR (net)	Trades/yr
<i>Low (monthly)</i>	Dual momentum [†]	42.8%	−51.3%	0.917	9.7
	SMA 200 crossover	33.1%	−64.6%	0.723	7.6
<i>Medium (daily)</i>	RSI(14) > 50	38.4%	−54.1%	0.804	44.0
	MACD crossover	25.2%	−55.0%	0.613	26.2
<i>Passive</i>	Buy & Hold	30.6%	−76.6%	0.618	0.1

[†]Upper-bound reference following Antonacci (2014); monthly rebalancing not directly comparable to intraday ML.

Every rule-based baseline in Table 11 outperforms the median model variant at its own frequency band (Table 10): RSI(14) reaches $SR = 0.804$ at 44 trades per year and MACD 0.613 at 26, while the median hourly model variant returns -1.32 . The dual momentum benchmark serves as an upper-bound reference; its superior SR reflects lower turnover (9.7 trades/year) rather than superior signal quality. Cross-section momentum strategies (24 configurations across 7 lookback periods and 3 portfolio sizes) similarly fail, with only 4 of 24 achieving $SR > 0$ (see Figures 5 and 6).

5.6. Ablation Analysis: Which VALID Items Matter Most?

Table 12 presents the ablation results.

V4 (CPCV/PBO) is the single most important item. Its removal causes FPR to increase from 0% to 27%.

6. Monte Carlo False Positive Analysis

6.1. Motivation and Definitions

The empirical results in Section 5 demonstrate failure modes on real data, but a natural question arises: how often would standard validation tools

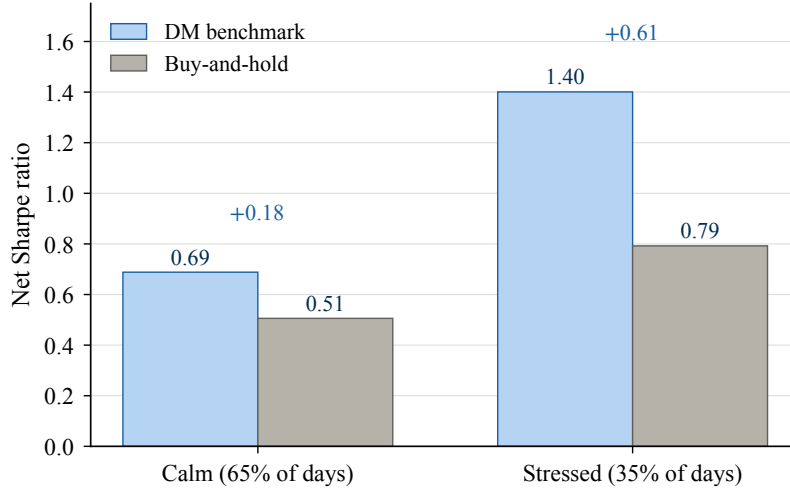


Figure 5: Regime-conditional performance under a two-state Gaussian HMM on 21-day rolling volatility. The momentum benchmark earns $SR = 1.40$ in stressed regimes (35% of evaluation days) versus 0.79 for buy-and-hold, and 0.69 versus 0.51 in calm periods: the spread widens from +0.18 to +0.61 under stress. No regime-conditional Sharpe ratio was computed for the model variants, so none is shown.

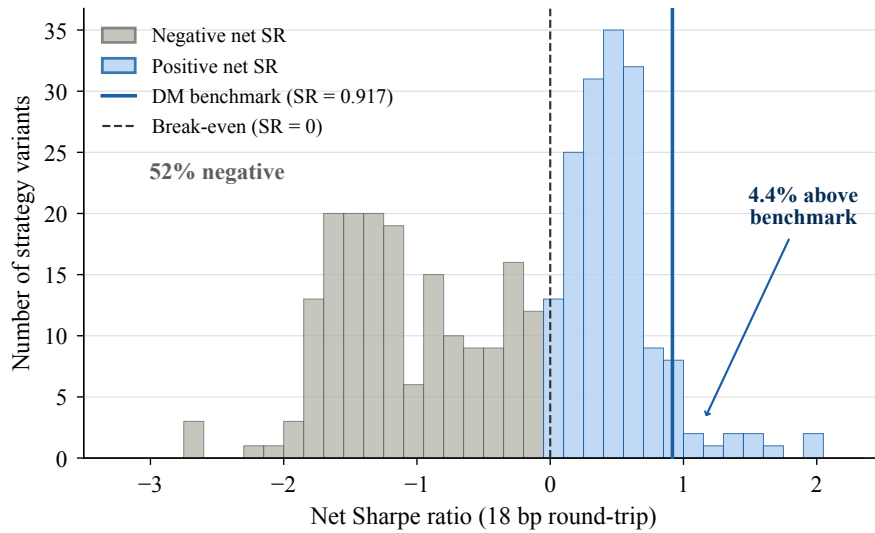


Figure 6: Distribution of net Sharpe ratios across the 340 strategy variants at 18 bp round-trip costs. Gray bars: negative net Sharpe (52%). Blue line: the DM benchmark ($SR = 0.917$), which 4.4% of variants exceed.

Table 12: Ablation: false positive rates by VALID item removal.

Configuration	Items Removed	FPR	95% Wilson CI
Full VALID	None	0.0%	[0.0%, 1.9%]
No CPCV/PBO	−V4	27.0%	[21.3%, 33.5%]
No balancing	−V2	0.0%	[0.0%, 5.8%]
No cost adj.	−V7	0.0%	[0.0%, 1.9%]
No permutation	−V6	0.0%	[0.0%, 1.9%]

478 produce false positives on data with *no embedded signal*? We address this
 479 through Monte Carlo simulation.

480 We define a **false positive** as a null (signal-free) dataset on which a given
 481 validation criterion incorrectly indicates the presence of a tradable signal.
 482 Specifically, for each criterion C and null dataset i , let $\text{FP}_i = \mathbf{1}[C_i \text{ passes}]$.
 483 The false positive rate is:

$$\text{FPR}(C) = \frac{1}{n} \sum_{i=1}^n \text{FP}_i \quad (6)$$

484 with 95% Wilson confidence intervals (Wilson, 1927). We evaluate five
 485 criteria: $\text{AUC} > 0.55$, permutation test $p < 0.05$, $\text{PBO} < 0.20$, $\text{net SR} > 0$, and
 486 the conjunction of all VALID items.

487 6.2. Synthetic Data Generation

488 We generate $n = 200$ synthetic BTC price series by bootstrapping (Efron
 489 and Tibshirani, 1993) daily log-returns (2017–2026, 3,132 observations per
 490 series). The bootstrap preserves the marginal distribution of returns (mean,
 491 variance, skewness, kurtosis) while destroying temporal dependence, ensuring
 492 that any predictive signal detected by the ML pipeline is spurious by con-
 493 struction. We use simple i.i.d. resampling rather than block or stationary
 494 bootstrap (Politis and Romano, 1994), as our goal is to eliminate all temporal
 495 structure. The choice of $n = 200$ yields Wilson CI half-widths of $\pm 6\%$ at
 496 $\text{FPR} = 27\%$, providing sufficient precision to distinguish informative from
 497 uninformative validation criteria.

498 6.3. Pipeline Application

499 Each synthetic series passes through the identical pipeline used for real
 500 data: (a) compute 17 price-derived technical features (returns, volatility,

momentum indicators); (b) generate Triple Barrier labels with CUSUM filtering; (c) train CatBoost with class-balanced sample weights; (d) evaluate via CPCV ($N = 6$, $k = 2$, yielding 15 train-test combinations); (e) record AUC, PBO, $\text{Var}(\text{SR}_{\text{IS}})$, permutation p -value, and net SR (18 bp round-trip cost). The pipeline is deterministic given the input data; no hyperparameter tuning is performed within the Monte Carlo loop.

6.4. Results

Mean AUC across 200 null iterations is 0.506 ± 0.070 , confirming that the pipeline produces near-chance classification on signal-free data. However, individual criteria show dramatically different false positive rates (Table 13; Equation 6). The 27% AUC-based FPR, combined with the audit finding that no empirical paper in our sample uses CPCV, suggests a substantial fraction of published positive results may reflect statistical artifacts. Critically, $\text{PBO} < 0.20$ alone achieves 0% FPR [0%, 1.9%], confirming its value as the primary overfitting diagnostic and justifying its central role in VALID (Item V4).

To confirm generalizability, we replicate the Monte Carlo analysis across three additional settings: ETH 1h, SOL 1h, and BTC daily (Table 13). AUC-based FPR ranges from 26.5% to 30.0% across all four settings, while PBO-based FPR remains 0.0% in every case. The consistency of these results across assets (BTC, ETH, SOL) and timeframes (hourly, daily) confirms that the findings are asset-invariant and timeframe-invariant: standard validation criteria produce false positives at similar rates regardless of the underlying data, and CPCV with PBO eliminates them universally.

Table 13: False positive rates on null data ($n = 200$ per setting). BTC 1h is the primary setting; three additional settings confirm generalizability. Column labels follow the original run names: all four settings were generated from daily-bar bootstrap series (838–919 events per series), a label mismatch recorded in the repository’s `REPRODUCE.md`. The conclusion is unaffected—every setting yields 0% under CPCV with PBO.

Criterion	BTC 1h	ETH 1h	SOL 1h	BTC daily
AUC > 0.55 alone	27.0%	30.0%	26.5%	27.0%
Permutation passed	22.0%	19.0%	18.0%	20.0%
PBO < 0.20 alone	0.0%	0.0%	0.0%	0.0%
Net SR > 0 alone	49.5%	52.0%	52.5%	49.5%
Full VALID	0.0%	0.0%	0.0%	0.0%

525 6.5. Multiple Testing Correction

526 The Monte Carlo analysis quantifies false positives on signal-free data. A
527 complementary question is what survives on the real corpus once the number
528 of trials is taken into account. Of the 340 variants, 15 (4.4%) exceed the
529 benchmark before correction, of which 10 are model variants; the remaining
530 five are benchmark entries of the cost-level and baseline blocks. We convert
531 each variant’s excess Sharpe to a t -statistic following Lo (2002) and apply
532 five corrections: Bonferroni ($N = 340$, $\alpha = 0.05$) leaves 9 survivors, Holm-
533 Bonferroni 9, Benjamini-Hochberg FDR 10, the Harvey et al. (2016) $t > 3.0$
534 threshold 10, and the Deflated Sharpe Ratio (Bailey and López de Prado,
535 2014) 0.

536 Cross-referencing the nine Bonferroni survivors against CPCV gives the
537 central result: *every one exhibits* $\text{PBO} = 1.0$, with t -statistics from 4.07 to
538 21.70 and AUC values between 0.47 and 0.64. All nine operate at the daily
539 timeframe. The Deflated Sharpe Ratio, which accounts for the expected
540 maximum Sharpe under N trials ($\mathbb{E}[\max] = 2.93$), rejects all 340: the best
541 observed Sharpe of 1.98 falls below the noise threshold. Classical corrections
542 control familywise or false-discovery error under independence, but they can-
543 not detect backtest overfitting—a strategy can satisfy $t > 21$ and $\text{PBO} = 1.0$
544 simultaneously. This is the corpus-level form of the disconnect documented
545 in Section 5.4.2, and it is why VALID places CPCV with PBO (Item V4) in
546 Stage 1 rather than treating multiple-testing correction as sufficient.

547 7. Discussion

548 7.1. Implications for the Field

549 Our findings have four implications. First, the crypto ML literature likely
550 contains a substantial false positive rate—across our 340 strategy variants,
551 52% produce negative net Sharpe ratios after realistic transaction costs, and
552 15 variants (4.4%) exceed the simple momentum benchmark ($\text{SR} = 0.917$),
553 of which 10 are model variants; the remaining five are benchmark entries
554 of the cost-level and baseline blocks. Second, our Monte Carlo analysis of
555 the $\text{Var}(\text{SR}_{\text{IS}})$ null distribution provides evidence consistent with Witzany’s
556 (2021) theoretical critique of PBO. Third, the statistical-economic disconnect
557 is not a failure of existing tools but a gap in how they are applied. Fourth, the
558 VALID framework addresses a genuine gap—no existing reporting standard
559 covers financial-ML-specific pitfalls.

560 *7.1.1. Adoption Barriers*

561 The VALID framework imposes computational and reporting overhead
562 that may deter adoption, particularly for research groups without access to
563 high-performance computing. CPCV with $\binom{6}{2} = 15$ combinations requires 15
564 full model training runs per configuration, and Monte Carlo analysis with
565 200 iterations multiplies this further. We view this cost as appropriate: if a
566 validation procedure is too expensive to run, the strategy search itself may be
567 too expansive to be trustworthy. Nonetheless, we acknowledge that pragmatic
568 compromises—such as reducing N or using fewer Monte Carlo iterations—may
569 be necessary for large-scale strategy searches, provided authors disclose the
570 reduced scope.

571 *7.1.2. Relationship to Publication Incentives*

572 The failure modes documented in this paper are not primarily technical
573 failures but incentive failures. Journals and conferences reward novelty and
574 positive results; negative results—such as our finding that 52% of strategy
575 variants produce negative net returns—are difficult to publish. The VALID
576 framework cannot resolve this structural problem, but by mandating disclosure
577 of class distributions, cost-adjusted performance, and baseline comparisons,
578 it raises the evidentiary bar for claiming positive results. We note that
579 analogous reporting standards in clinical research (CONSORT; Moher et al.,
580 2010; Collins et al., 2015) have measurably improved methodological quality
581 over time, though compliance remains imperfect.

582 *7.1.3. Generalizability Beyond Cryptocurrency*

583 While our empirical validation focuses on cryptocurrency markets, the
584 three failure modes—directional prediction bias, statistical-economic discon-
585 nect, and cost illusion—are not crypto-specific. Bull bias arises whenever
586 training labels are imbalanced, which occurs in any trending asset class. The
587 statistical-economic disconnect exists wherever statistical validation tools
588 lack economic grounding. Cost illusion affects any strategy with non-trivial
589 turnover. We expect VALID to be applicable, with minor parameter ad-
590 justments, to equity factor models, FX strategies, and commodity trading
591 algorithms.

592 *7.2. Limitations*

593 Our analysis is subject to several limitations. First, we evaluate three
594 cryptocurrency assets (BTC, ETH, SOL); extending to a broader universe

including equities, foreign exchange, or fixed income would strengthen generalizability claims. Second, the VALID framework has not been validated through a formal expert consensus process such as a Delphi study, which is standard for clinical reporting guidelines (TRIPOD, CONSORT). Third, the momentum benchmark operates at monthly rebalancing frequency while ML strategies trade hourly to daily; although we provide frequency-matched comparisons using RSI and MACD (Table 11), the cross-frequency comparison warrants caution. Fourth, our deep learning evaluation covers five architectures but omits Temporal Fusion Transformers (Lim et al., 2021) and graph neural networks, which may exploit cross-asset dependencies not captured here. Fifth, the literature audit was conducted by a single coder without independent inter-rater reliability assessment, introducing potential coding bias. Sixth, the audit covers only published and indexed papers, creating survivorship bias—unpublished negative results are absent by construction. Seventh, our 18 bp fixed cost model, while conservative for hourly and daily frequencies, does not capture the volume-dependent slippage dynamics relevant to sub-minute execution.

7.3. Future Work

Three extensions merit investigation. First, applying VALID retroactively to highly cited papers would quantify how many published results survive rigorous validation. Second, extending the framework to reinforcement learning strategies—which face additional challenges in reward shaping and environment non-stationarity—would broaden applicability. Third, developing an automated VALID compliance tool, analogous to existing TRIPOD adherence checkers, would lower adoption barriers.

8. Conclusion

We propose VALID which, to our knowledge, is the first checklist-style protocol for financial ML that pairs binary statistical and economic gates in a fixed reporting order. Through a systematic audit of 80 papers, 340 strategy variants across three assets and four timeframes, 800-iteration Monte Carlo analysis across four asset-timeframe settings, and ablation testing, we demonstrate three systematic failure modes: bull bias (90–97% for tree models, 58–86% across seven deep learning architectures), statistical-economic disconnect, and cost illusion (net Sharpe falling monotonically with trading frequency), with 52% of all tested variants producing negative returns after

costs and 15 (4.4%) exceeding a simple momentum benchmark, of which 10 are model variants. Nine variants survive Bonferroni correction—all with $PBO = 1.0$ —and none survives the Deflated Sharpe Ratio. Our $\text{Var}(SR_{IS})$ null-distribution analysis (real 0.012, null 95th percentile 0.328) provides evidence consistent with Witzany’s (2021) PBO critique. The VALID framework’s 12 items would establish minimum reporting standards for a field that currently lacks them.

In a field where positive results are rewarded and negative results are invisible, the most important validation is the discipline to accept that a statistically detectable signal is not the same as a tradable edge.

CRediT Authorship Contribution Statement

Jaewook Kim: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The author declares that there is no known competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

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Data Availability

Bitcoin, Ethereum, and Solana OHLCV data were obtained from Binance and Bybit public APIs via the CCXT library. Macro indicators were obtained from Yahoo Finance. The literature audit coding sheet (`audit/literature_audit_80.csv`), the 340-variant corpus

659 (results/reference/variants_340.csv) and the scripts that recompute
660 every published corpus figure (scripts/) are in the public repository; Monte
661 Carlo per-iteration results are provided in supplementary materials. Source
662 code is available at <https://github.com/orcajae/valid-framework>.

663 Appendix A: VALID Self-Assessment

664 To demonstrate the framework’s application, Table 14 applies VALID to
665 the present study. This self-assessment serves as a worked example for future
666 adopters.

667 V5 is partial because the absolute and null-relative flatness criteria disagree
668 for our own pipeline (Section 5.4.2); V8 is partial because all corpus variants
669 are evaluated at a single 18 bp cost level, so we report a frequency ladder
670 rather than a cost sweep of our own strategies; V10 remains partial: the
671 regime analysis covers two volatility states on three cryptocurrency assets.
672 This self-assessment was revised downward after auditing our own submission
673 against VALID.

Table 14: VALID self-assessment applied to this study.

#	Item	Status	Evidence
V1	Class distribution reported	✓	Table 7: 90–97% long
V2	Balanced/unbalanced comparison	✓	Tables 7, 8
V3	Temporal splitting only	✓	CPCV with purge & embargo
V4	CPCV with PBO	✓	$N = 6$, $k = 2$, 15 combinations
V5	Var(SR _{IS}) reported	~	Table 9: criteria disagree
V6	Permutation tests	✓	$p = 0.000$; $n = 100$ (meets the ≥100 floor; 200 in three of four settings)
V7	Net performance with costs	✓	18 bp RT; Table 10
V8	Cost sensitivity analysis	~	Single 18 bp level; Section 5.5
V9	Simple baselines	✓	RSI, MACD, SMA in Table 11
V10	Bear market evaluation	△	Figure 5: two regimes only
V11	Trade frequency reported	✓	Trades/yr in Table 11
V12	Code available	✓	GitHub repository linked
Total		9/12	3 partial

674 **Appendix B: Supplementary Materials**

675 Supplementary materials include: (A) complete 340-variant perfor-
676 mance results; (B) VALID framework implementation code; (C) the
677 complete 80-paper coding sheet, available in the public repository
678 (audit/literature_audit_80.csv); (D) Monte Carlo simulation details
679 and per-iteration results (200 iterations).

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